

E-commerce Customer Insights

ABC & RFM analysis

Data Analyst portfolio Project

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Project overview

Background

This project focuses on analyzing customer behavior based on e-commerce platform serving both B2B and B2C customers. The project is based on two years worth of companies raw customer data (22.2.2023-21.2.2025) and the goal was to transform the raw data into actionable business insights and actions.

The Business Problem

The e-commerce platform lacked a structured view of their customer base.

- which of the customers are the most critical and how we ensure that they stay?
- How can we identify and react to customer churn?

Key Highlights

Dataset: 10 000
transactions

Duration: 2 years of data

Output: Customer
segmentation (B2B & B2C)

This project is based on the aliazimieco Dataset. (2023). Retrieved from
[<https://www.kaggle.com/datasets/aliazimieco/rfm-dataset>].

Data Cleaning & Processing



End-to-end Python project

Tech stack: Python, Pandas, NumPy & Matplotlib

Cleaning & Validation

Initial data: 10 000 rows → Final Data: 9 301 rows.

checked for duplicate customer IDs and removed negative spend values, and age outliers (kept range 18-79)

Feature Engineering

Currency Conversion: Unified 4 currencies to EUR using ECB exchange rates (as of 19.3.2026)

B2B & B2C Segmentation to separate dataframes: Classified based on payment method (B2B if credit) and total transaction (>10k€ B2B)

Classification

ABC Analysis: Ranked by cumulative revenue (A:80%, B:15% & C: 5%)

RFM Scoring: Calculated Recency by defining a snapshot date (next day from the most recent purchase of the data 22.2.2025) and compared every customers most recent purchase to that date. Used `pd.qcut` to create 1-5 scoring for recency scores. Defined F scores by using `np.select`.

RFM segmentation: Created conditions to divide customers to 3 different groups in all 3 ABC classes.

Active high frequency: last purchase in under 104 days and over 4 purchases.

Active low frequency: last purchase in under 104 days and under 4 purchases.

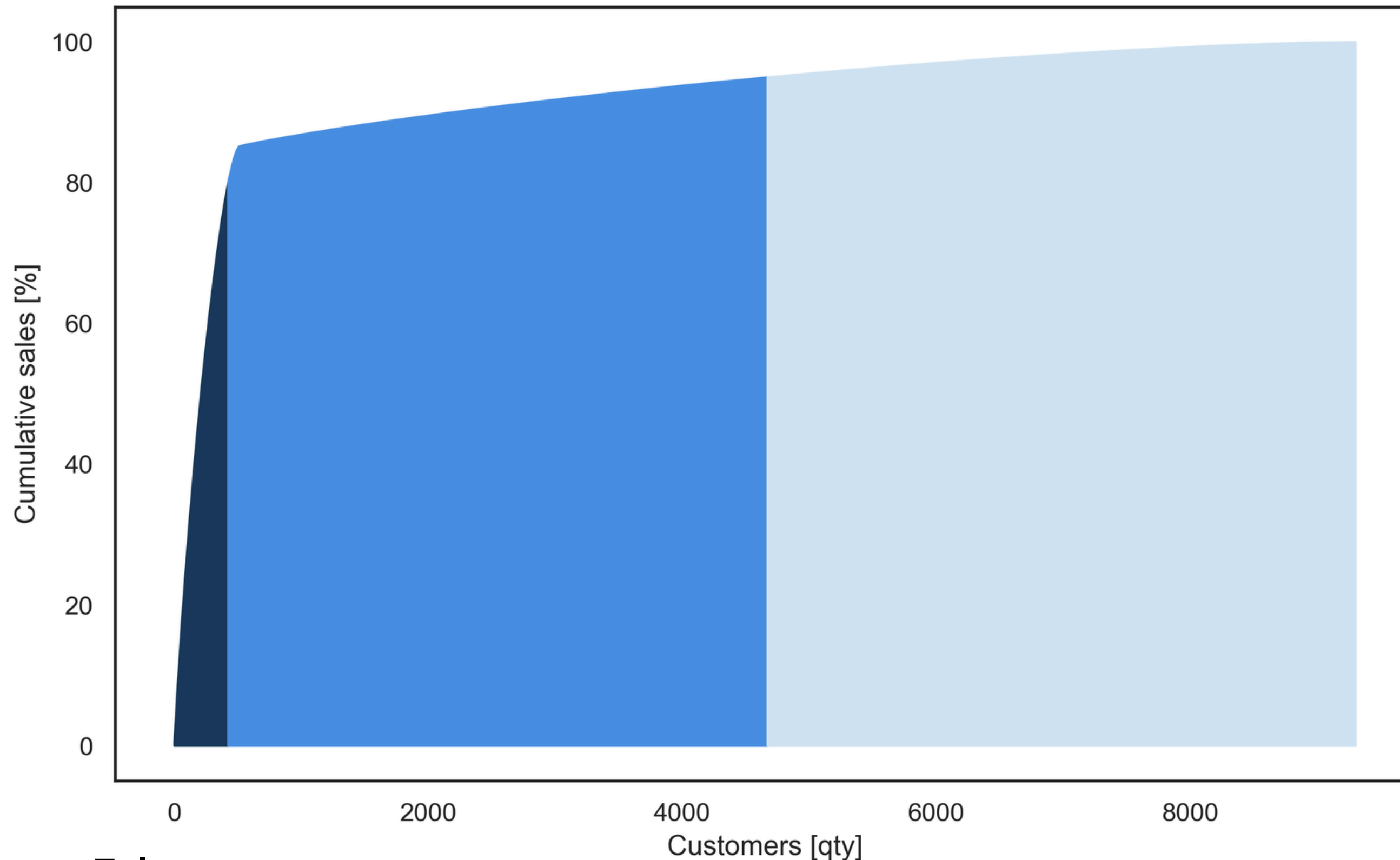
Passive: last purchase over 104 days ago.

Defined RFM categories using `np.select`.

Visualization

Used Matplotlib to plot Cumulative Revenue Curve for ABC analysis and stacked bar charts for RFM.

ABC -analysis



Classes:

A

423 customers (5%)

All are B2B.

B

4248 customers (45%)

68% B2C & 32% B2B

C

4630 customers (50%)

71% B2C & 29% B2B

Takeaways:

- Just **5%** of total customers generate **80%** of total revenue and all of these are business clients
- Classes B & C represent a massive **95%** portion of the customer base yet generate only **20%** of total revenue.

RFM -analysis

Next we want to take a deeper look into customer behavior in ABC groups. For this analysis we use 3 metrics.

R

Recency

How many days since
last order?

Active:

If less than 149 days

Passive:

If over 149 days

F

Frequency

How many transactions
the customer has made?

High F:

More than 4

Low F:

Less than 4

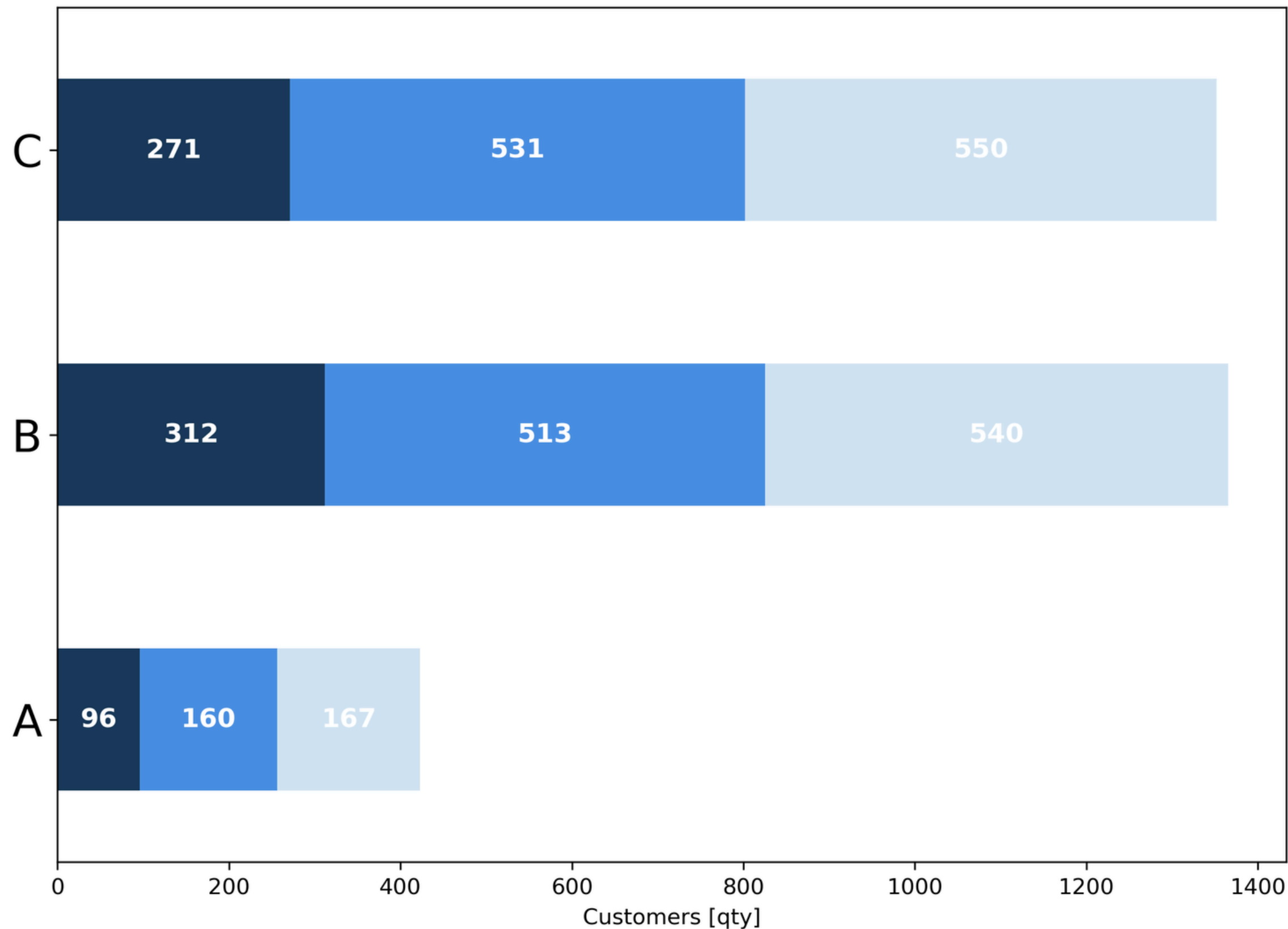
M

Monetary

How much total value
has the customer
brought?

ABC classes

RFM -analysis B2B



Active customers with high frequency

- From **A** class the most important customers.
- **B & C** smaller but loyal business customers.
- **22%** of B2B customers.

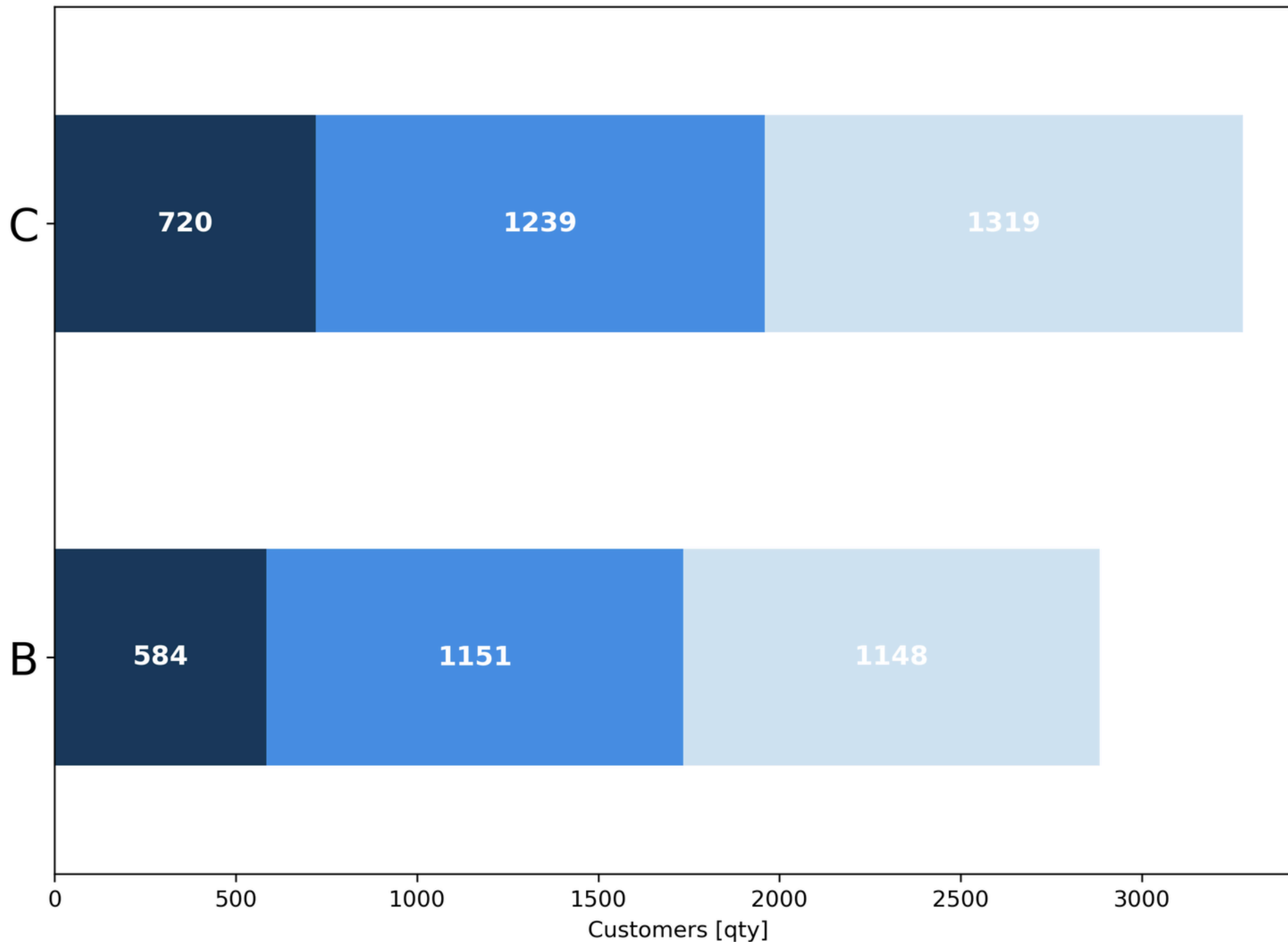
Active customers with low frequency

- New customers testing our platform for the first time.
- **38%** of B2B customers.

Passive customers

- Customers that haven't used our platform in over 104 days.
- **40%** of B2B customers are passive.

RFM -analysis B2C



Active customers with high frequency

- Loyal customers who buy often
- represent **21%** of B2C customers

Active customers with low frequency

- New customers testing our platform for the first time.
- represent **39%** of B2C customers

Passive customers

- Customers that haven't used our platform in over 104 days.
- **40%** of **B2C** customers are passive.

Insights & Actions

Maximizing Loyalty Among Active Users

We need to keep our active customers with high frequency. These are the most important customers of our platform.

Action:

We should send **VIP-benefits** for A-class customers and make a **loyalty program** for B & C so these clients would stay with us and not change to a competitor.

Our platform is leaking

40% of customers are passive and 38% are new, this means that our platform is losing more customers than it is getting new ones.

Action:

- For passive customers we should start a **Win-back program** which sends personalized email containing special offer for next order and purchase recommendations based on the clients buying history.
- After first purchase in our platform we should send email containing purchase recommendations based on the first order and advertisements about our loyalty program.

By implementing these actions, we can recapture passive revenue and increase customer lifetime value